QP CODE: 24900099

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION (2024)

First Semester

Discipline Specific Core Course - MG1DSCEVS100 - ENVIRONMENTAL BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Short answers

Answer 10 questions Each question carries 1 mark

[Learning Domain] [CO No(s)]

1. **Name any four nutrient cycles [K] [2]**
   Nitrogen cycle, Oxygen cycle, Carbon cycle, Sulphur cycle
2. **Define Poikilotherms. [K] [2]**
   Poikilotherms are organisms with variable body temperatures that change based on the temperature on their surroundings
3. **What is food chain? [K] [2]**
   A Food chain is the sequence of organisms that eat each other in an ecosystem, and the transfer of energy from one organism to another.
4. **How is net primary productivity is different from gross primary productivity? [An] [2]**
   Gross primary productivity is the total amount of energy captured producers in an ecosystem. Net primary productivity is the amount of energy remaining after accounting for plant respiration and maintenance.
5. **What is bio magnification? [U] [2]**
   Bio magnification is the process by which a substance, such as pollutants or pesticides, increase in concentration as it moves up the food chain.
6. **Define synecology. [U] [1]**
   Synecology (community ecology), is the study of how species interact with each other in a community.
7. **What are the important actions we can take to protect the Environment? [A] [1]**

Advocate for Environmental friendly and sustainable practices/ Promote reforestation activities /Protect environmental resources.

8. **What are the three main types of biodiversity? [U] [3]**
   Ecosystem Diversity, Species Diversity, Genetic Diversity
9. **Give two examples for Keystone species. [K] [3]**
   Sea stars, Bees

10. What you meant by biodiversity hot-spots? Give two examples. [U] [3]

Biodiversity hot-spots are regions where high levels of biodiversity that are threatened by human activities and other threats. Eg: Western ghats, Sundaland/Himalaya.

11. Name two important scopes of environmental studies [K] [1]

Addressing Environmental challenges, Understanding the Environment

12. Analyze how human activities contribute to the disruption of the Environment. [An] [1]

Habitat destruction /Pollution/Deforestation /Misuse of natural resources.

(10 x1 = 10)

PART B

Short Essay Type Questions

Answer 6 Questions. Each question carries 5 marks.

13. Explain biotic and abiotic components of an ecosystem with examples. [U] [1]

Biotic components of an ecosystem are all living things in the ecosystem, include producers, consumers, and decomposers.

- **Producers**: Also known as autotroph, these organisms produce food through photosynthesis. Eg: green plants and cyanobacteria.
- **Consumers**: Also known as heterotrophs, these organisms depend on other autotrophs for food. Consumers are further classified into primary, secondary, tertiary consumers. Eg: Tiger, elephant, human beings
- **Decomposer**: Also known as saprophytes, these organisms break down dead and decaying organic matter into simpler molecules. Eg: fungi

The abiotic components of an ecosystem are the nonliving parts of the ecosystem including air, water, soil, minerals, sunlight, temperature, nutrients, wind etc

14 What is pyramid of biomass? [U] [2]

- Biomass is the total number mass of living organisms in a specific unit of area or volume of habitat
- A pyramid of biomass is a graphical representation of the total mass of living organisms at each trophic level in an ecosystem
- It shows the relationship between biomass and tropic levels and the flow of energy from producers to consumers
- In terrestrial ecosystem the pyramid of biomass is usually upright, but in marine environment its often inverted

15 Explain in-situ conservation methods with suitable example? [U] [3]

In-situ conservation is the reservation of species in their natural habitat to protect the environment and ensure the sustainability of the ecosystem.

In-situ conservation is often used for wild and endangers species to avoid the risk of extinction. Eg:- National parks:- small reserves with well-defined boundaries where human activities like gracing, forestry and cultivation are prohibited.

Biosphere reserves: protected areas that conserve ecosystem and natural habitat

Wildlife sanctuaries: areas that conserve wildlife.

16. Differentiate between tropical grasslands and temperate grassland? [An] [2]

- **Tropical grasslands** are warm,grassy ecosystems with distinct wet and dry seasons that are found in Africa,Asia,Australia and South America .
- It receive 50 to 130 cm of rainfall per year.
- Tropical grasslands have a greater density of woody shrubs and trees than temperate grasslands.

- **Temperate grasslands** are found in mid latitude , usually in the interiors of continents they are located in; North America- prairies, South America- pampas Asia and Europe- steppes
- Temperate grasslands have distinct seasons with cold winter and warm summer.
- They receive low to moderate annual precipitation.

17. How is primary productivity measured? [E] [2]

- Primary productivity is the rate at which organisms make their own food from inorganic sources.
- It is usually measured by quantifying the amount of oxygen produced or carbon dioxide assimilated.
- Primary productivity is usually expressed as grams of organic carbon per unit area per unit time.
- Primary productivity can be classified into two types, Gross Primary productivity(GPP) and Net Primary Productivity (NPP).

Net Primary Productivity(NPP) = Gross Primary Productivity(NPP) - Respiration

18. Explain the multidisciplinary nature of Environmental Studies? [An] [1]

- Environmental studies is a multidisciplinary field of study because it draws from different disciplines to understand the environment and the impact on human.
- This subject incorporates the knowledge and methods from many disciplines including, Biology, Chemistry, Physics, Geology, Engineering, Sociology, Economics and more.
- Environmental studies is an applied science that seeks practical answers to making human civilization sustainable on Environmental resources.
- This subject concerned with understanding how humans interact with the environment and how to solve complex problems.

19. How does the human activities affect the nutrient cycling? [U] [2]

- Humans activities like deforestation and burning of fuels increase the amount of carbon dioxide in the atmosphere.
- Activities like using fertilizers can increase the amount nitrogen in soil and affects the nitrogen cycle.
- Soil pollution is a major loss pathway for phosphorus in many land ecosystems.
- Increasing number of automobiles can create an imbalance in sulphur content in our environment.

20. Give four adaptation of desert animals. [K] [2]

- Wide feet to prevent them for sinking in the sand.
- Producing less urine or having a higher tolerance for salt.
- Humps are present to store fat so they can go months without food.
- Long eyelashes and hairy ears to keep sand and sun out of their eyes and ears.

(6X5=30)

PART C
Essay Type Questions

Answer any one question. Question carries 10 marks

21. What is the nitrogen cycle? What are the steps involved in biological nitrogen fixation. [U] [2]

- Nitrogen cycle is a biogeochemical process that moves nitrogen between the atmosphere, land and living things.
- Most of the nitrogen in the earth atmosphere is in the form of nitrogen gas (N2), which is not directly usable by living organisms.

- **Nitrogen fixation:** Bacteria or lightening convert nitrogen gas into ammonia, a form of nitrogen that plant can use.
- **Nitrification:** Nitrifying bacteria convert ammonia into nitrates, which plant can absorb.
- **Denitrification:** Denitrifying bacteria convert nitrates and nitrites back into nitrogen gas, which is released into the atmosphere.
- **Ammonification:** When plants and animals die, fungi break down their remains into ammonia.
- The nitrogen cycle is essential for maintaining the balance of ecosystems.
- Human activities like using nitrogen fertilizers, burning fossil fuels, and releasing nitrogen into waste water can alter the nitrogen cycle and harm the environment and human health.

22. Define Red data book. How it influence the biodiversity conservation? Explain. [A] [3]

- Red data book is a public document that records endangered, rare, and vulnerable species of plant, animal, and fungi.
- It is used to help identify and protect species that are at risk of extinction.
- The red data book was first issued by the special survival commission of the IUCN IN 1964.
- The red data book uses colour coded information sheets to organize species by their risk of extinction
- **Black:** Species that are confirmed extinct.
- **Red:** Species that are endangered
- **Amber:** Species that are vulnerable
- **Green:** Species that wear once endangered but are recovering
- **White:** Rare species
- **Grey**:Species that are vulnerable,endangered or rare, but there isn't enough information to classify them
- The red data book classifies species into categories like threatened, not threatened, and unknown.
- It also provide information on the population of particular species and helps to estimate the risk of species becoming extinct.
- Red data book provides guidelines for protecting endangered species. (1X10=10)